DiDi: Full Evaluation of DAM4SAM and SAMURAI

This report evaluates existing predictions only. Neither tracker was rerun, and no future ground truth (GT) was supplied to either model. The evaluation covers all 180 sequences and 274,882 frames, with 360 prediction trajectories across the two models. The current batch contains 179 sequences; the previously completed leaves results are reused.

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The PDF includes the full analysis, every sequence-level result, two statistical charts, and six case contact sheets. Videos remain separate files in the offline gallery; they are not embedded in this PDF. DAM denotes DAM4SAM and SAM denotes SAMURAI in compact tables. Scores are rounded to four decimal places; full precision is retained in all-sequences.csv.

1. Overall results

Model	Quality Q	Accuracy A	Robustness R
DAM4SAM	0.6860	0.7173	0.9436
SAMURAI	0.6793	0.7207	0.9319

The mean Q difference, SAMURAI minus DAM4SAM, is **-0.0066** (-0.66 percentage points). DAM4SAM has higher Q on 102 sequences and SAMURAI on 78; there are no ties within 1e-6.

A paired sequence bootstrap over these 180 sequences (10,000 resamples; seed 20261027) gives a 95% percentile interval of [-0.0288, +0.0159] for the mean Q difference. This describes exploratory variation within the current fixed collection. Source sequences in DiDi may be correlated, so this interval does not establish general superiority on independent data.

The median paired Q difference is -0.0015, and the correlation between the models' sequence-level Q scores is 0.755. Mean differences, median differences, and win counts answer different questions. Large failures on individual sequences must also be examined to interpret the aggregate gap.

Figure 1: All-sequence Q scores and paired differences. See the landscape figure plates at the end.

2. Shared protocol and comparability

Evaluation and annotation selection

- The main metrics follow move_didi_results.py and didi-workspace/baseline_experiment.yaml from the fixed DAM4SAM source. Both models are scored with VOT Toolkit 0.7.1. Predicted masks are converted to enclosing bounding boxes; the original bytes of the active groundtruth.txt files are preserved.
- An annotation-selection issue was discovered and corrected. The original archive contains 90 groundtruth_old.txt backup files: 86 from LaSOT and four from GOT-10k. The default VOT loader prioritizes groundtruth_*.txt over groundtruth.txt and therefore loads these backups as an object named old. Active and backup annotations differ on 5,955 frames, all involving active Special(0) versus a backup rectangle. No valid target geometry was rewritten during this evaluation.
- The final derived data view preserves the backup files outside the sequence loader's directory namespace. For all 180 sequences, checks confirm that the loaded object comes from the active groundtruth.txt, has the correct length, and has the same SHA256 as the original archive member. Model outputs, original data, and source annotation files were

not modified. Earlier default-loader results remain in audit-unfiltered-loader and are not mixed into the final statistics.

- With backups incorrectly loaded, the earlier Q scores were 0.6816 for DAM4SAM and 0.6751 for SAMURAI. The main table reports the final scores using active GT. The change reflects annotation selection during scoring, not improved inference or parameter tuning.

Metric definitions and initialization

- Q uses burnin=0, ignore_unknown=False, and weighted=False, with equal weight per sequence in the final average. The original conversion script replaces the first-frame prediction with Special(1). Its scoring follows the installed toolkit; this report does not manually exclude the first frame to raise scores. A measures overlap on successful tracking frames, subject to the initialization detail below. R defines success as IoU > 0, not IoU >= 0.5.

- The installed implementation has a first-frame edge case. In gather_overlaps, Special(1) enters the initialization branch before the later threshold branch. With burnin=0, that frame is not removed and does not subsequently pass through the threshold branch. Q and A can therefore include a zero-overlap initialization entry when first-frame GT is valid. Native scoring is preserved. Frame-level diagnostics explicitly exclude the input initialization frame.

- Both models use the same original DiDi sequences, the SAM2.1 Large checkpoint, the provided first-frame segmentation mask, and no intermediate GT correction. All initialization masks were checked again during evaluation. This is a controlled mask-initialized SAMURAI comparison; it does not reproduce the box initialization used by its default demo.

- Q, A, and R from the official analysis CLI were cross-checked against aggregation through the same installed toolkit functions. All six final values agree to within 1e-12; see official-crosscheck.json. Aggregate Q must not be inferred by multiplying aggregate A and aggregate R.

Compact evaluation view and runtime differences

- To avoid transferring all 28 GB of images again, scoring uses a compact data view. Original GT and tag bytes are preserved, and derived sequence metadata adds length, width, and height. Lengths and first-frame dimensions were checked for all 180 sequences. GOT-10k Val 000014, leaves, and birds2 also retain all images; for both models, their Q/A/R inputs and valid-frame indices match between the complete directories and the compact view. No synthetic images replace real frames.

- DAM4SAM runs with PyTorch 2.1.0 and SAMURAI with PyTorch 2.3.1. Assigned GPUs include A100, A100 MIG, L40S, and H200. SAMURAI uses a CPU image cache, with sampled normalized image tensors checked for exact equality against the native loader. Prediction quality can be compared for these two fixed implementations. Timing logs are not a matched-environment speed benchmark or an ablation that changes only memory management.

The released annotations contain 260,975 RECTANGLE frames, 7,696 SPECIAL frames, and 6,211 MASK frames. Native mask GT was not silently replaced with box GT.

Most unavailable intervals are encoded as Special(0) and skipped under this toolkit's protocol. Q therefore does not fully evaluate absence decisions or false positives during those intervals. A nonempty prediction on a frame without valid GT cannot automatically be labeled an error. Recovery evaluation should start when valid annotation becomes available again.

GT shape also affects comparisons between source groups. The same predicted box can overlap a thin or sparse mask GT less than it overlaps a box GT. Differences between VOT-origin and LaSOT-origin averages cannot be interpreted directly as differences in scene difficulty. Representative cases require review of target location and the native predicted masks.

Comparison with published numbers

Table 2 of the DAM4SAM paper reports Q/A/R of 0.694/0.727/0.944 for SAM2.1++ and 0.680/0.722/0.930 for SAMURAI. These are contextual references, not evidence of exact reproduction. Annotation release, code versions, and the shared initialization protocol remain relevant. Source: DAM4SAM paper, Table 2.

Our DAM4SAM Q is 0.0080 lower than that table, and SAMURAI Q is 0.0007 lower. These discrepancies have not been fully localized. The main table supports a unified comparison of the two fixed implementations, not a claim that every published number has been exactly reproduced.

3. Where both models struggle

- Both models have whole-sequence Q below 0.5 on **30 sequences**, and below 0.3 on **seven sequences**. These are screening thresholds chosen for this analysis, not official dataset categories.
- Both have IoU < 0.2 for at least 10 consecutive valid, target-present frames on **48 sequences**, and for at least 30 such frames on **32 sequences**.
- Excluding initialization and frames without valid target-present GT leaves 266,984 diagnostic frames. Both models have low overlap on 21,565 of them, or 8.1%.
- A stricter candidate pattern for delayed discovery requires both models to have mean IoU > 0.5 over the preceding up-to-30-frame valid prefix, followed by at least 30 consecutive shared IoU < 0.2 frames. There are **two sequences and two intervals**. This establishes a trajectory pattern, not the cause of an identity error or memory contamination.
- Consecutive low-overlap intervals do not bridge unknown-GT or target-absent frames. Recovery must be assessed from subsequent curves. Pixel overlap between a native predicted mask and box GT is not segmentation accuracy; native mask IoU has a direct segmentation interpretation only where mask GT is available.

Figure 2: Sequences with the highest shared low-overlap fractions. See the landscape figure plates at the end.

The following grouping uses original source-name labels. It is descriptive and does not represent the complete official test set of any source dataset.

Source	Sequences	DAM Q	SAM Q	Both Q < 0.5
GOT-10k	4	0.7771	0.7500	0
UTB180	56	0.7736	0.7754	2
VOT2022-ST	20	0.4139	0.4112	14
LaSOT	86	0.7195	0.6996	9
VOT2020-LT	6	0.6784	0.7290	0
VOT2020-ST	1	0.1705	0.3285	1
VOT2022-LT	7	0.3780	0.3950	4

The largest differences between models are also retained, so that analysis does not focus exclusively on shared failures.

Sequence	DAM Q	SAM Q	Delta Q
LaSOT_car-17	0.7945	0.1141	-0.6804
UTB180_Video_0002	0.2234	0.8906	+0.6672
UTB180_Video_0066	0.1978	0.8618	+0.6639
LaSOT_motorcycle-18	0.0899	0.7378	+0.6479
LaSOT_hat-1	0.6802	0.0915	-0.5887
UTB180_Video_0103	0.8107	0.2376	-0.5731
LaSOT_kite-10	0.6965	0.2007	-0.4958
LaSOT_turtle-8	0.6399	0.1753	-0.4646

Sequence	DAM Q	SAM Q	Delta Q
LaSOT_monkey-4	0.7773	0.3379	-0.4394
UTB180_Video_0140	0.7782	0.3635	-0.4147

All 30 shared difficult sequences are listed below, sorted by the larger of the two Q scores in ascending order. Shared-low percentage refers to frames where both IoUs are below 0.2 among valid target-present diagnostic frames; longest run is a contiguous shared-low interval in frames. The full list is retained rather than showing only selected stories.

Sequence	DAM Q	SAM Q	Shared-low %	Longest run	Video
VOT2022-ST_flamingo1	0.1595	0.1051	70.0%	744	Video
LaSOT_train-1	0.0535	0.1816	53.2%	540	Video
VOT2022-ST_birds2	0.2339	0.2203	32.9%	14	Video
VOT2022-ST_zebrafish1	0.2386	0.2449	29.6%	15	Video
VOT2022-LT_hand	0.2454	0.2166	35.6%	42	Video
LaSOT_pig-18	0.1821	0.2655	64.2%	1036	Video
VOT2022-ST_rabbit	0.2741	0.2459	48.4%	50	Video
VOT2022-ST_hand2	0.1964	0.3023	49.3%	19	Video
VOT2022-ST_ants1	0.2540	0.3261	10.8%	17	Video
VOT2020-ST_drone_flip	0.1705	0.3285	20.5%	20	Video
LaSOT_hand-9	0.3165	0.3380	33.6%	30	Video
VOT2022-LT_dancingshoe	0.2065	0.3456	36.3%	330	Video
UTB180_Video_0120	0.3465	0.3231	42.9%	95	Video
VOT2022-LT_goldfish	0.2955	0.3602	37.1%	537	Video
VOT2022-LT_skiing	0.2825	0.3619	41.8%	55	Video
VOT2022-ST_marathon	0.3624	0.3548	3.0%	1	Video
LaSOT_robot-19	0.2777	0.3674	36.7%	494	Video
VOT2022-ST_drone1	0.3713	0.3704	14.7%	52	Video
LaSOT_robot-1	0.3997	0.4007	51.8%	1312	Video
VOT2022-ST_birds1	0.3952	0.4020	1.5%	2	Video
LaSOT_tank-9	0.4242	0.4161	48.6%	430	Video
VOT2022-ST_bolt1	0.4278	0.4240	0.0%	0	Video
VOT2022-ST_basketball	0.4303	0.4298	0.1%	1	Video
LaSOT_zebra-16	0.2322	0.4342	29.7%	252	Video
VOT2022-ST_polo	0.4403	0.4396	0.0%	0	Video
LaSOT_surfboard-12	0.4572	0.4280	4.9%	13	Video
UTB180_Video_0107	0.4726	0.3486	34.7%	93	Video
VOT2022-ST_handball1	0.4814	0.3574	4.0%	6	Video

Sequence	DAM Q	SAM Q	Shared-low %	Longest run	Video
VOT2022-ST_fish2	0.4835	0.4795	0.0%	0	Video
LaSOT_microphone-2	0.4930	0.4664	10.9%	58	Video

4. Representative cases: observations and limits

Quantitative statistics cover all 180 sequences. Qualitative review uses initialization frames, frames before and after candidate failures, key frames within low-overlap intervals, and local video segments. It is not a manual frame-by-frame review of all 274,882 frames. Native mask overlays use yellow for DAM4SAM, blue for SAMURAI, and cyan for active GT. The reused leaves visualization uses pink predictions on both sides. Screening thresholds retrieve cases; they do not affect model predictions.

Switching to the other shoe: VOT2022-LT_dancingshoe

DAM Q = 0.2065; SAM Q = 0.3456. Shared-low fraction: 36.3%; longest shared-low run: 330 frames.

The target is one designated shoe on a dancer. Around frame 1858, both models' overlap with that shoe drops sharply. At frames 1894, 1930, and 1960, cyan GT still identifies the designated shoe, while both predictions lie near the other shoe. These frames have valid GT; missing annotation does not explain their low scores.

This is a candidate case of identity confusion between similar object parts. It is a strong starting point for delayed correction: after tracking has moved to the other shoe, supply one correct mask and test whether the original shoe can be tracked consistently afterward. Current results contain no memory read/write traces or counterfactual interventions, so they do not identify a causal memory channel.

Full video | Representative clip

Figure 3: Switching to the other shoe: VOT2022-LT_dancingshoe / Key frames. See the landscape figure plates at the end.

Persistent confusion between similar boars: LaSOT_pig-18

DAM Q = 0.1821; SAM Q = 0.2655. Shared-low fraction: 64.2%; longest shared-low run: 1,036 frames.

The first frame specifies one wild boar. After two animals approach, overlap, and change relative positions, predictions separate from the cyan target box. Frame 1280 makes the mismatch clear: GT is on the animal to the left, while both nonempty predictions mainly cover the other animal to the right. Later key frames also show disagreement with the designated target.

This is closer to persistently treating the wrong object as the target than to simply producing empty predictions, and it provides an external look-alike distractor case. Grass occlusion, visible parts, and box boundaries also affect IoU. Identity judgments should use initialization and the full video, not a low-score curve alone.

Full video | Representative clip

Figure 4: Persistent confusion between similar boars: LaSOT_pig-18 / Key frames. See the landscape figure plates at the end.

Confusion within a zebra group: LaSOT_zebra-16

DAM Q = 0.2322; SAM Q = 0.4342. Shared-low fraction: 29.7%; longest shared-low run: 252 frames.

Later in the sequence, the designated zebra approaches neighboring zebras. At frames 811, 937, and 1092, cyan GT and yellow/blue predictions lie on different animals. Some predictions also cover multiple neighboring instances. Both models drift; this is more than a predicted outline being slightly smaller than the GT box.

The case complements the shoe example by showing external same-category distractors and mixed-instance predictions. Single-target GT does not assign instance IDs to the other zebras. Rigorous identity-switch counts

would require manual instance annotation. Current evidence confirms disagreement with the designated target, not an exact count of identity switches.

Full video | Representative clip

Figure 5: Confusion within a zebra group: LaSOT_zebra-16 / Key frames. See the landscape figure plates at the end.

Low scores from expanding target extent: VOT2022-ST_flamingo1

DAM Q = 0.1595; SAM Q = 0.1051. Shared-low fraction: 70.0%; longest shared-low run: 744 frames.

The initialization target and cyan GT specify the flamingo's head region. In many later frames, both models' predictions expand to the neck, body, or whole bird, creating a predicted enclosing box much larger than the head GT. Similar expansion appears from part of a train to multiple carriages in train-1, and from hand to forearm in hand-9.

This is an important counterexample: low Q for both models does not necessarily mean a switch to another object. Segmentation granularity bias, part-target preservation, and evaluation geometry may all contribute. Treat these as a separate category and control; do not force every low-score case into a memory-contamination explanation.

Full video | Representative clip

Figure 6: Low scores from expanding target extent: VOT2022-ST_flamingo1 / Key frames. See the landscape figure plates at the end.

Control where SAMURAI is much better: UTB180_Video_0002

DAM Q = 0.2234; SAM Q = 0.8906. Shared-low fraction: 0.0%; longest shared-low run: zero frames.

The target fish crosses or overlaps a larger fish/shark. At frames 337, 367, and 397, SAMURAI's predicted box still overlaps cyan GT strongly, whereas DAM4SAM's prediction occupies another region or is clearly separated from the target.

This reverse control avoids selecting only SAMURAI failures. Overall averages cannot replace sequence-level analysis. A new repair method must also be tested for damage to cases that already track correctly.

Full video | Representative clip

Figure 7: Control where SAMURAI is much better: UTB180_Video_0002 / Key frames. See the landscape figure plates at the end.

Divergence after valid annotation resumes: LaSOT_car-17

DAM Q = 0.7945; SAM Q = 0.1141. Shared-low fraction: 2.6%; longest shared-low run: 18 frames.

Part of this sequence has active GT marked unavailable; that interval is not counted as a failure here. At frame 2145, when valid annotation is available again, DAM4SAM overlaps the target box strongly while SAMURAI remains displaced. Frame 1977, which also has valid GT, shows a substantial difference earlier.

This is a control for different recovery behavior. It motivates examining the two trackers' states around loss and recovery, but does not establish whether motion gating, memory updates, or another implementation detail causes the difference. Scoring must first exclude intervals without valid GT.

Full video | Representative clip

Figure 8: Divergence after valid annotation resumes: LaSOT_car-17 / Key frames. See the landscape figure plates at the end.

5. Implications for the research direction

The most promising question is how to recover when correct evidence arrives after the tracker has already lost the intended object. These unmodified trajectories identify natural segments worth testing. They do not establish which memory stores contain an error, or whether clearing pointers, removing spatial memory, or replaying frames will help. Without model confidence, memory read/write logs, and explicit IDs for additional objects, low IoU alone does not prove a high-confidence identity switch.

Start with dancingshoe: the designated shoe and similar distractor shoe are clear, and shared drift can be reviewed at specific frames. Then use pig-18 and zebra-16 to test confusion with external same-category instances. Keep flamingo1, hand-9, and train-1 in a separate target-extent expansion category rather than merging them into a single identity-contamination claim.

The next experiment should fix a development subset of manually reviewed natural events. At the same delayed-correction frame, give both models equivalent correct masks and compare correction of the current mask only, clearing suspect history, restoring a trusted state, and selective repair. Keep predictions submitted before correction unchanged; evaluate current and future outputs, degradation on normal segments, and additional cost. These interventions have not been run in this report.

Annotation conditions must be explicit. Most DiDi sequences provide frame-level boxes and only a first-frame mask. A later box is not a precise GT mask. A small development set can receive manually annotated correction masks, or a separate, shared box-correction protocol can be defined. The source of external correction evidence during training and testing must be distinguished from automatic error detection. An oracle diagnostic is not evidence of a GT-free automatic method.

DiDi is a stress test for distractor scenarios. Its observed failures can guide development, but later main results need independent development/test splits and broader datasets. Sequences already used to select cases are exploratory development data; after repeated tuning, they must not be presented as unseen test evidence.

6. Files and job records

- The offline index.html gallery contains all videos, metrics, curves, and filters. Keep its videos directory beside it for playback. Opening the PDF does not require a running local web server.
- Frame-level metrics are in evaluation-results/frames. Protocol, annotation-integrity, and compact-view equivalence records include input-integrity.json and metadata-equivalence-check.json. The official toolkit output is workspace/analysis/dual-full.json under evaluation-results.
- Server-side evaluation inputs, logs, and archives are in /home/wang2895/didi-evaluation-20261027-v1/. The 179 batch video originals are in /home/wang2895/didi-dual-20261025-v1/videos/. The local gallery adds the earlier leaves video for a total of 180. This evaluation used CPU jobs and added no GPU inference tasks.
- Job 7219894 was replaced during full-archive transfer. Job 7224183 records the default loader's backup-annotation results. Job 7225695 rescored the existing predictions against active annotations. These were not tracker failures and did not rerun inference. Details are retained in evaluation-job-receipt.json.

Appendix: All 180 sequence-level comparisons

Frames is the sequence length. Delta Q is SAMURAI minus DAM4SAM. Shared-low percentage and longest run use the valid target-present diagnostic frames defined in Section 3, excluding initialization. Video links refer to local companion files and may require opening the offline gallery in a browser. All sequence names, numeric cells, and video destinations below are preserved from the Chinese report.

Sequence	Frames	DAM Q	SAM Q	Delta Q	Shared-low %	Longest run	Video
GOT-10k_GOT-10k_Val_000014	100	0.7856	0.7273	-0.0582	4.2%	4	Video
GOT-10k_GOT-10k_Val_000082	90	0.7967	0.7208	-0.0759	0.0%	0	Video
GOT-10k_GOT-10k_Val_000126	101	0.8824	0.8854	+0.0030	0.0%	0	Video
GOT-10k_GOT-10k_Val_000175	271	0.6435	0.6666	+0.0231	0.4%	1	Video
UTB180_Video_0002	484	0.2234	0.8906	+0.6672	0.0%	0	Video
UTB180_Video_0015	105	0.9385	0.9370	-0.0015	0.0%	0	Video
UTB180_Video_0016	105	0.9032	0.8946	-0.0087	0.0%	0	Video
UTB180_Video_0024	220	0.8745	0.8632	-0.0114	0.0%	0	Video
UTB180_Video_0030	308	0.7601	0.7633	+0.0032	0.0%	0	Video
UTB180_Video_0034	176	0.6189	0.6443	+0.0254	0.0%	0	Video
UTB180_Video_0042	105	0.8277	0.8196	-0.0081	0.0%	0	Video
UTB180_Video_0046	350	0.8699	0.8679	-0.0020	0.0%	0	Video
UTB180_Video_0047	451	0.7826	0.7676	-0.0150	2.4%	6	Video
UTB180_Video_0051	588	0.8213	0.8292	+0.0080	1.7%	8	Video
UTB180_Video_0056	804	0.7680	0.7596	-0.0084	0.5%	4	Video
UTB180_Video_0063	324	0.9225	0.9211	-0.0013	0.0%	0	Video
UTB180_Video_0065	309	0.8483	0.8386	-0.0097	0.0%	0	Video
UTB180_Video_0066	405	0.1978	0.8618	+0.6639	4.2%	15	Video
UTB180_Video_0067	391	0.9200	0.9095	-0.0105	0.0%	0	Video
UTB180_Video_0070	441	0.8287	0.8442	+0.0155	0.0%	0	Video
UTB180_Video_0082	250	0.6479	0.6289	-0.0190	6.4%	9	Video
UTB180_Video_0087	178	0.9374	0.9209	-0.0165	0.0%	0	Video
UTB180_Video_0088	462	0.5722	0.6915	+0.1193	0.0%	0	Video
UTB180_Video_0093	318	0.8487	0.8348	-0.0139	0.0%	0	Video
UTB180_Video_0099	130	0.8068	0.7987	-0.0080	0.0%	0	Video
UTB180_Video_0100	125	0.8405	0.8469	+0.0064	0.0%	0	Video
UTB180_Video_0101	120	0.8136	0.8040	-0.0096	0.0%	0	Video
UTB180_Video_0103	213	0.8107	0.2376	-0.5731	0.0%	0	Video
UTB180_Video_0105	232	0.8264	0.8282	+0.0018	0.0%	0	Video
UTB180_Video_0106	361	0.8216	0.8170	-0.0046	0.0%	0	Video

Sequence	Frames	DAM Q	SAM Q	Delta Q	Shared-low %	Longest run	Video
UTB180_Video_0107	269	0.4726	0.3486	-0.1240	34.7%	93	Video
UTB180_Video_0110	237	0.8725	0.8324	-0.0401	0.0%	0	Video
UTB180_Video_0111	415	0.7722	0.8149	+0.0427	0.2%	1	Video
UTB180_Video_0112	390	0.7332	0.7743	+0.0411	0.0%	0	Video
UTB180_Video_0113	386	0.7907	0.7873	-0.0034	0.0%	0	Video
UTB180_Video_0114	284	0.8731	0.8834	+0.0104	0.0%	0	Video
UTB180_Video_0117	213	0.7669	0.7728	+0.0059	0.0%	0	Video
UTB180_Video_0118	232	0.8651	0.8491	-0.0160	0.0%	0	Video
UTB180_Video_0119	232	0.8457	0.6624	-0.1834	0.0%	0	Video
UTB180_Video_0120	232	0.3465	0.3231	-0.0233	42.9%	95	Video
UTB180_Video_0122	429	0.6669	0.7128	+0.0459	0.5%	2	Video
UTB180_Video_0123	220	0.8128	0.8136	+0.0008	2.7%	6	Video
UTB180_Video_0125	451	0.9193	0.9174	-0.0019	0.0%	0	Video
UTB180_Video_0126	316	0.6496	0.6558	+0.0063	0.0%	0	Video
UTB180_Video_0127	316	0.7325	0.7096	-0.0229	0.3%	1	Video
UTB180_Video_0130	334	0.7103	0.7094	-0.0009	0.0%	0	Video
UTB180_Video_0132	503	0.8283	0.8260	-0.0024	0.0%	0	Video
UTB180_Video_0133	363	0.7649	0.7554	-0.0095	0.0%	0	Video
UTB180_Video_0134	338	0.8814	0.8777	-0.0037	0.0%	0	Video
UTB180_Video_0137	391	0.8138	0.8019	-0.0119	0.0%	0	Video
UTB180_Video_0138	391	0.8695	0.8626	-0.0069	0.0%	0	Video
UTB180_Video_0139	386	0.8324	0.8395	+0.0071	0.0%	0	Video
UTB180_Video_0140	386	0.7782	0.3635	-0.4147	0.5%	1	Video
UTB180_Video_0142	363	0.8409	0.8324	-0.0085	0.0%	0	Video
UTB180_Video_0143	471	0.8612	0.8289	-0.0323	0.0%	0	Video
UTB180_Video_0151	266	0.7876	0.8081	+0.0205	0.0%	0	Video
UTB180_Video_0152	292	0.9395	0.9441	+0.0046	0.0%	0	Video
UTB180_Video_0153	387	0.6255	0.6465	+0.0210	17.6%	44	Video
UTB180_Video_0156	213	0.8757	0.8695	-0.0062	0.0%	0	Video
UTB180_Video_0160	370	0.7643	0.7776	+0.0133	0.0%	0	Video
VOT2022-ST_ants1	325	0.2540	0.3261	+0.0721	10.8%	17	Video
VOT2022-ST_basketball	725	0.4303	0.4298	-0.0005	0.1%	1	Video
VOT2022-ST_birds1	339	0.3952	0.4020	+0.0068	1.5%	2	Video
VOT2022-ST_birds2	80	0.2339	0.2203	-0.0136	32.9%	14	Video

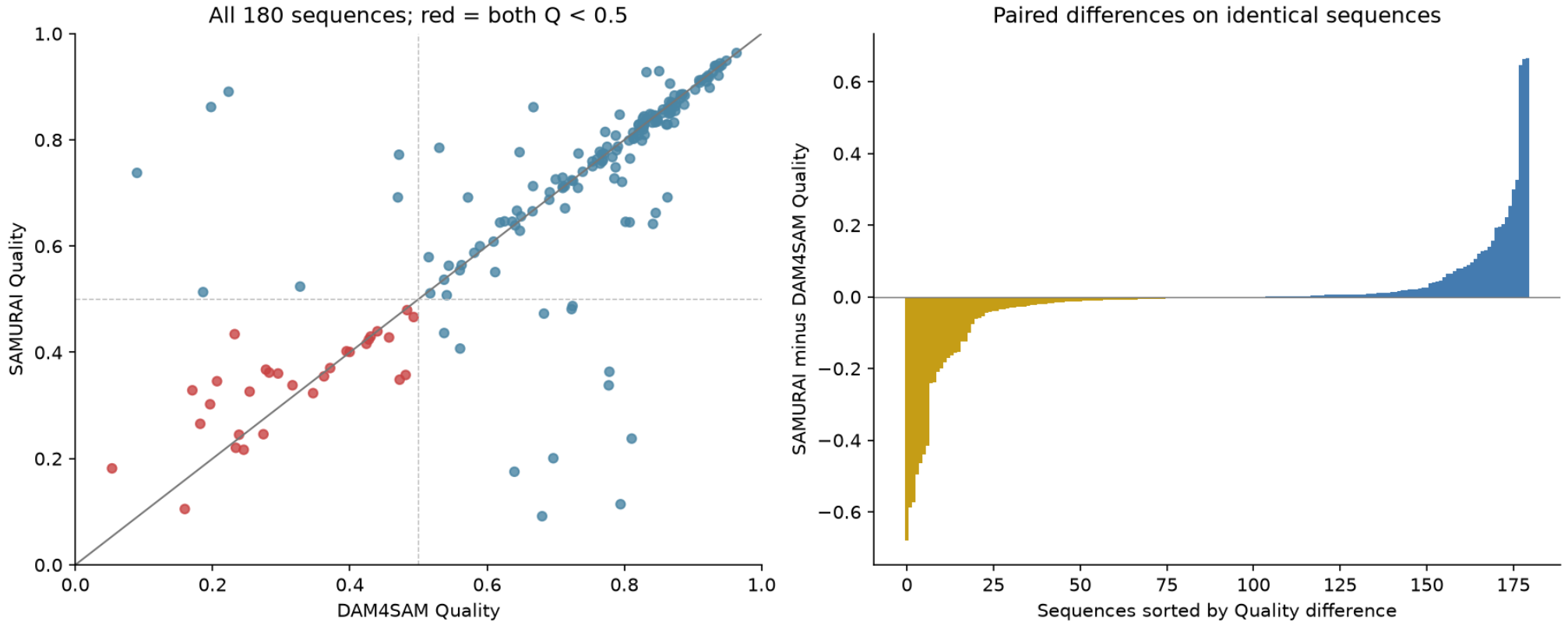
Sequence	Frames	DAM Q	SAM Q	Delta Q	Shared-low %	Longest run	Video
VOT2022-ST_bolt1	350	0.4278	0.4240	-0.0037	0.0%	0	Video
VOT2022-ST_crabs1	160	0.5172	0.5111	-0.0060	1.9%	1	Video
VOT2022-ST_drone1	355	0.3713	0.3704	-0.0009	14.7%	52	Video
VOT2022-ST_fish2	310	0.4835	0.4795	-0.0040	0.0%	0	Video
VOT2022-ST_flamingo1	1377	0.1595	0.1051	-0.0544	70.0%	744	Video
VOT2022-ST_hand2	145	0.1964	0.3023	+0.1058	49.3%	19	Video
VOT2022-ST_handball1	377	0.4814	0.3574	-0.1240	4.0%	6	Video
VOT2022-ST_lamb	100	0.5629	0.5642	+0.0014	0.0%	0	Video
VOT2022-ST_leaves	63	0.5813	0.5874	+0.0061	3.2%	2	Video
VOT2022-ST_marathon	101	0.3624	0.3548	-0.0076	3.0%	1	Video
VOT2022-ST_monkey	101	0.6663	0.6655	-0.0007	0.0%	0	Video
VOT2022-ST_polo	111	0.4403	0.4396	-0.0007	0.0%	0	Video
VOT2022-ST_rabbit	158	0.2741	0.2459	-0.0282	48.4%	50	Video
VOT2022-ST_soccer1	392	0.6411	0.6390	-0.0022	2.3%	6	Video
VOT2022-ST_soccer2	129	0.5603	0.5545	-0.0058	0.0%	0	Video
VOT2022-ST_zebrafish1	400	0.2386	0.2449	+0.0063	29.6%	15	Video
LaSOT_basketball-1	1088	0.7133	0.7125	-0.0008	0.1%	1	Video
LaSOT_bird-15	1575	0.6474	0.7767	+0.1294	0.0%	0	Video
LaSOT_bird-2	4101	0.7933	0.8474	+0.0541	0.0%	0	Video
LaSOT_bottle-12	2400	0.8882	0.8837	-0.0045	0.0%	1	Video
LaSOT_bus-19	1930	0.8686	0.8705	+0.0019	0.1%	2	Video
LaSOT_car-17	2251	0.7945	0.1141	-0.6804	2.6%	18	Video
LaSOT_cattle-13	2321	0.7235	0.7239	+0.0004	0.3%	6	Video
LaSOT_cattle-2	2461	0.8551	0.8501	-0.0050	0.0%	1	Video
LaSOT_chameleon-20	1650	0.5150	0.5792	+0.0642	0.2%	3	Video
LaSOT_coin-18	1100	0.9320	0.9392	+0.0073	0.1%	1	Video
LaSOT_coin-3	1020	0.8323	0.9273	+0.0950	0.0%	0	Video
LaSOT_cup-1	1433	0.8629	0.8290	-0.0340	0.0%	0	Video
LaSOT_deer-10	1411	0.6095	0.6084	-0.0011	7.9%	40	Video
LaSOT_deer-8	2911	0.8083	0.7647	-0.0436	0.9%	17	Video
LaSOT_dog-1	1168	0.8270	0.8406	+0.0136	0.3%	2	Video
LaSOT_dog-19	1514	0.8855	0.8858	+0.0003	0.1%	1	Video
LaSOT_drone-15	1362	0.7245	0.4872	-0.2374	6.5%	14	Video
LaSOT_drone-7	2817	0.7538	0.7502	-0.0036	1.0%	8	Video

Sequence	Frames	DAM Q	SAM Q	Delta Q	Shared-low %	Longest run	Video
LaSOT_electricfan-10	1601	0.9488	0.9489	+0.0001	0.0%	0	Video
LaSOT_electricfan-18	1863	0.8629	0.6917	-0.1712	0.3%	3	Video
LaSOT_elephant-1	1889	0.8019	0.6456	-0.1563	6.4%	36	Video
LaSOT_elephant-12	1053	0.8634	0.8603	-0.0031	1.4%	10	Video
LaSOT_flag-3	1904	0.8671	0.8557	-0.0114	0.1%	1	Video
LaSOT_gametarget-1	2251	0.8743	0.8545	-0.0198	0.0%	0	Video
LaSOT_gametarget-2	1834	0.8675	0.8505	-0.0170	0.3%	2	Video
LaSOT_gecko-5	1251	0.5304	0.7850	+0.2546	0.0%	0	Video
LaSOT_goldfish-3	2375	0.5896	0.5998	+0.0102	6.1%	21	Video
LaSOT_goldfish-8	2250	0.1862	0.5134	+0.3272	10.6%	227	Video
LaSOT_gorilla-9	1839	0.8455	0.8330	-0.0124	0.2%	3	Video
LaSOT_guitar-10	1890	0.7137	0.6712	-0.0425	5.8%	35	Video
LaSOT_hand-9	1733	0.3165	0.3380	+0.0216	33.6%	30	Video
LaSOT_hat-1	3486	0.6802	0.0915	-0.5887	7.3%	95	Video
LaSOT_hat-5	8000	0.5443	0.5631	+0.0188	30.2%	1382	Video
LaSOT_helmet-19	1408	0.6913	0.7013	+0.0099	0.0%	0	Video
LaSOT_kangaroo-5	3119	0.4700	0.6917	+0.2216	3.6%	63	Video
LaSOT_kite-10	3309	0.6965	0.2007	-0.4958	0.5%	6	Video
LaSOT_kite-4	3223	0.7704	0.7737	+0.0034	0.6%	2	Video
LaSOT_kite-6	3774	0.7750	0.7869	+0.0120	1.1%	7	Video
LaSOT_licenseplate-12	2685	0.6367	0.6459	+0.0093	0.1%	2	Video
LaSOT_microphone-2	4017	0.4930	0.4664	-0.0266	10.9%	58	Video
LaSOT_monkey-3	1261	0.8292	0.8296	+0.0004	0.0%	0	Video
LaSOT_monkey-4	1435	0.7773	0.3379	-0.4394	2.8%	26	Video
LaSOT_motorcycle-18	2557	0.0899	0.7378	+0.6479	1.6%	6	Video
LaSOT_person-12	1990	0.7887	0.7805	-0.0082	0.0%	0	Video
LaSOT_person-5	2967	0.9635	0.9635	-0.0000	0.0%	0	Video
LaSOT_pig-18	2401	0.1821	0.2655	+0.0834	64.2%	1036	Video
LaSOT_pig-2	2671	0.9126	0.9125	-0.0001	0.0%	0	Video
LaSOT_pool-12	1801	0.9292	0.9277	-0.0015	0.0%	0	Video
LaSOT_pool-15	1338	0.9334	0.9395	+0.0061	0.3%	1	Video
LaSOT_pool-3	2256	0.8207	0.8279	+0.0072	0.5%	4	Video
LaSOT_rabbit-10	2098	0.8507	0.9293	+0.0785	0.2%	3	Video
LaSOT_racing-10	1508	0.8652	0.8715	+0.0063	0.0%	0	Video

Sequence	Frames	DAM Q	SAM Q	Delta Q	Shared-low %	Longest run	Video
LaSOT_racing-16	2186	0.8563	0.8570	+0.0007	0.1%	2	Video
LaSOT_robot-1	5609	0.3997	0.4007	+0.0010	51.8%	1312	Video
LaSOT_robot-19	5797	0.2777	0.3674	+0.0897	36.7%	494	Video
LaSOT_robot-5	2325	0.7232	0.4814	-0.2418	0.0%	1	Video
LaSOT_rubicCube-1	1800	0.9235	0.9181	-0.0054	0.1%	1	Video
LaSOT_rubicCube-19	2101	0.9092	0.9119	+0.0028	0.1%	1	Video
LaSOT_sepia-13	2756	0.9420	0.9404	-0.0016	0.0%	0	Video
LaSOT_shark-2	3455	0.3276	0.5238	+0.1962	0.5%	8	Video
LaSOT_sheep-5	1000	0.8298	0.8094	-0.0204	0.0%	0	Video
LaSOT_sheep-7	1256	0.9364	0.9372	+0.0008	0.1%	1	Video
LaSOT_sheep-9	1840	0.8656	0.8582	-0.0074	0.0%	0	Video
LaSOT_skateboard-8	4112	0.6118	0.5510	-0.0608	12.9%	484	Video
LaSOT_spider-20	9999	0.8667	0.9058	+0.0392	0.0%	0	Video
LaSOT_surfboard-12	3259	0.4572	0.4280	-0.0292	4.9%	13	Video
LaSOT_surfboard-5	4383	0.8209	0.8082	-0.0127	0.0%	1	Video
LaSOT_swing-10	2189	0.9244	0.8981	-0.0263	0.0%	0	Video
LaSOT_tank-16	3635	0.8452	0.8459	+0.0007	0.0%	0	Video
LaSOT_tank-9	1312	0.4242	0.4161	-0.0082	48.6%	430	Video
LaSOT_tiger-18	3107	0.8374	0.8485	+0.0111	0.0%	0	Video
LaSOT_tiger-6	2295	0.7394	0.7399	+0.0005	0.0%	0	Video
LaSOT_train-1	1150	0.0535	0.1816	+0.1281	53.2%	540	Video
LaSOT_train-11	1080	0.8258	0.7985	-0.0273	0.0%	0	Video
LaSOT_train-7	1425	0.8078	0.6448	-0.1629	0.9%	11	Video
LaSOT_truck-16	7225	0.8417	0.6417	-0.2000	3.6%	17	Video
LaSOT_turtle-8	1497	0.6399	0.1753	-0.4646	0.1%	1	Video
LaSOT_umbrella-17	1001	0.9099	0.9076	-0.0023	0.0%	0	Video
LaSOT_umbrella-19	2868	0.8873	0.8664	-0.0209	0.2%	2	Video
LaSOT_umbrella-9	1615	0.7687	0.7619	-0.0068	7.9%	112	Video
LaSOT_volleyball-1	1309	0.7253	0.7226	-0.0027	0.0%	0	Video
LaSOT_volleyball-13	2042	0.7533	0.7593	+0.0060	0.4%	1	Video
LaSOT_volleyball-18	2606	0.6829	0.4729	-0.2100	4.6%	38	Video
LaSOT_volleyball-19	1467	0.5607	0.4071	-0.1536	10.3%	34	Video
LaSOT_zebra-10	2008	0.6675	0.8615	+0.1939	0.8%	14	Video
LaSOT_zebra-16	1282	0.2322	0.4342	+0.2020	29.7%	252	Video

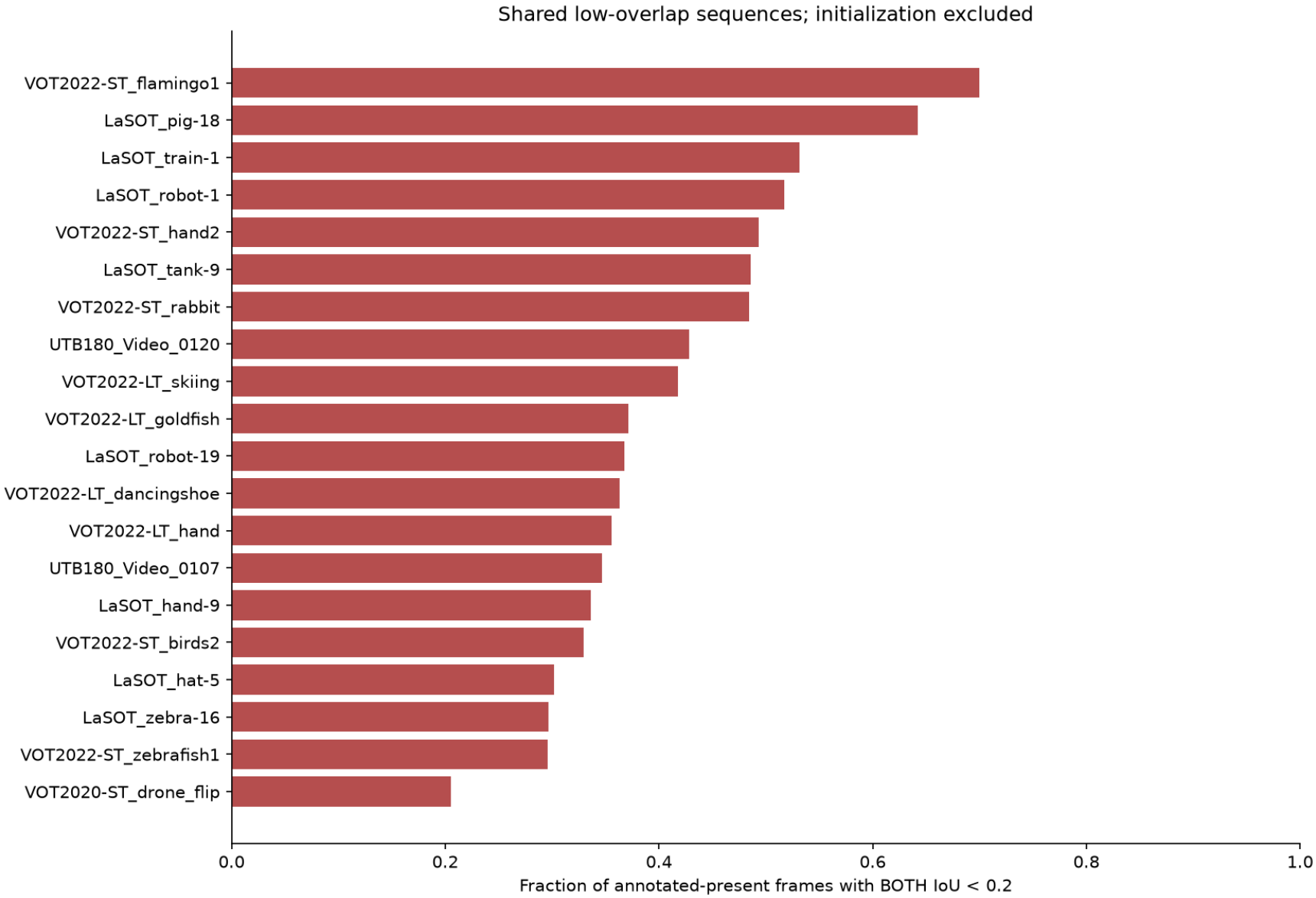
Sequence	Frames	DAM Q	SAM Q	Delta Q	Shared-low %	Longest run	Video
VOT2020-LT_car1	2629	0.7873	0.7483	-0.0390	10.4%	253	Video
VOT2020-LT_car3	1717	0.7101	0.7288	+0.0187	1.3%	4	Video
VOT2020-LT_group1	4873	0.7000	0.7256	+0.0256	0.0%	2	Video
VOT2020-LT_group2	2683	0.7104	0.7121	+0.0017	1.5%	37	Video
VOT2020-LT_group3	5527	0.6906	0.6873	-0.0034	0.0%	2	Video
VOT2020-LT_warmup	3961	0.4717	0.7720	+0.3003	0.1%	4	Video
VOT2020-ST_drone_flip	113	0.1705	0.3285	+0.1580	20.5%	20	Video
VOT2022-LT_dancingshoe	5451	0.2065	0.3456	+0.1391	36.3%	330	Video
VOT2022-LT_diving	1765	0.5413	0.5075	-0.0337	5.5%	41	Video
VOT2022-LT_goldfish	4520	0.2955	0.3602	+0.0647	37.1%	537	Video
VOT2022-LT_hand	1830	0.2454	0.2166	-0.0288	35.6%	42	Video
VOT2022-LT_run	1712	0.5375	0.5367	-0.0008	0.0%	0	Video
VOT2022-LT_sailing	1812	0.5375	0.4363	-0.1012	0.1%	1	Video
VOT2022-LT_skiing	3057	0.2825	0.3619	+0.0794	41.8%	55	Video

Figure 1 | All-sequence Q scores and paired differences



Computed from the final unified evaluation of all 180 sequences. Thresholds are diagnostic criteria, not official dataset categories.

Figure 2 | Sequences with the highest shared low-overlap fractions



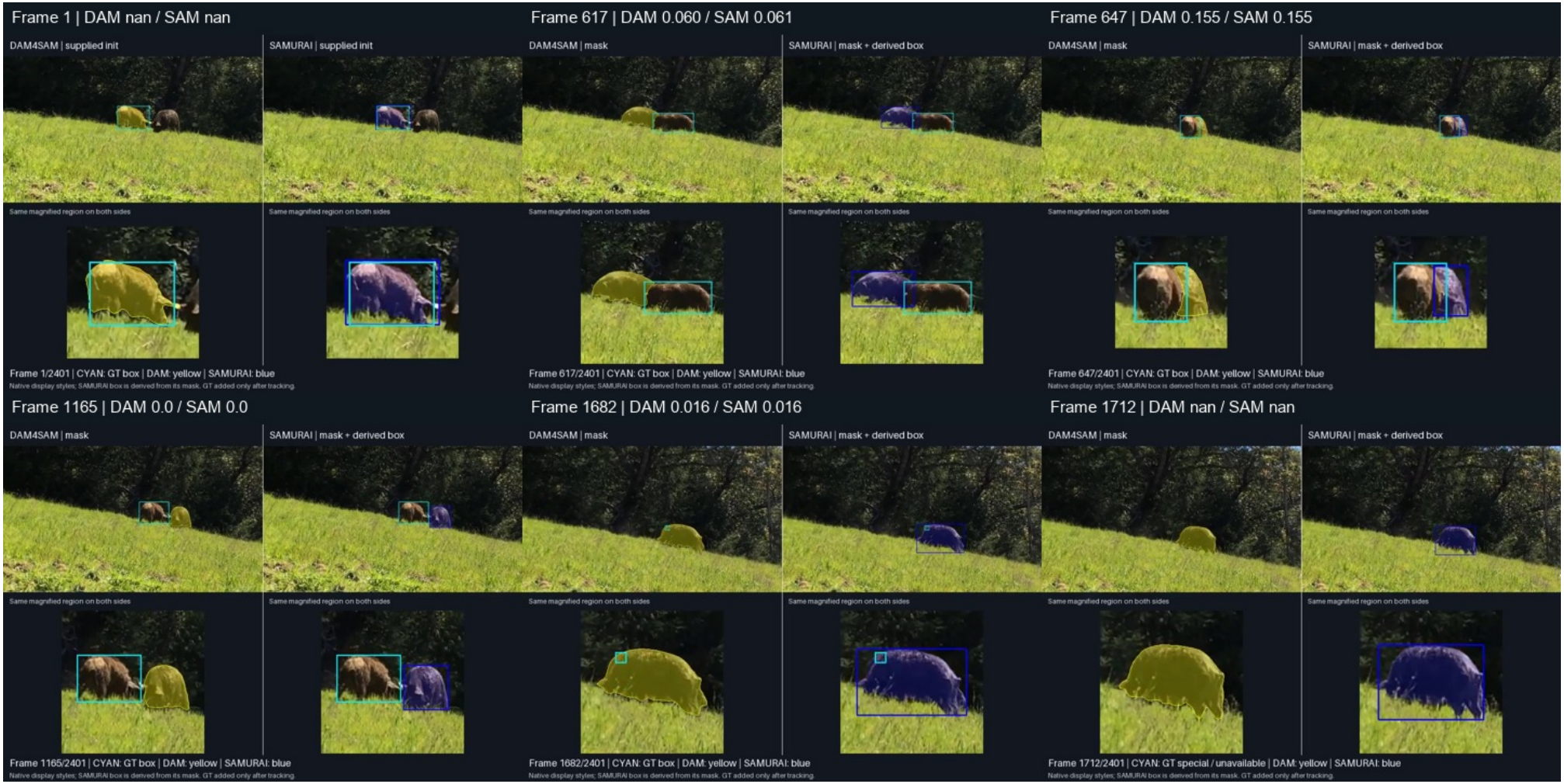
Computed from the final unified evaluation of all 180 sequences. Thresholds are diagnostic criteria, not official dataset categories.

Figure 3 | Switching to the other shoe: VOT2022-LT_dancingshoe / Key frames



Cyan: active GT. Yellow: DAM4SAM. Blue: SAMURAI. The panels retain native mask predictions; the full dynamic sequence is available in the companion gallery.

Figure 4 | Persistent confusion between similar boars: LaSOT_pig-18 / Key frames



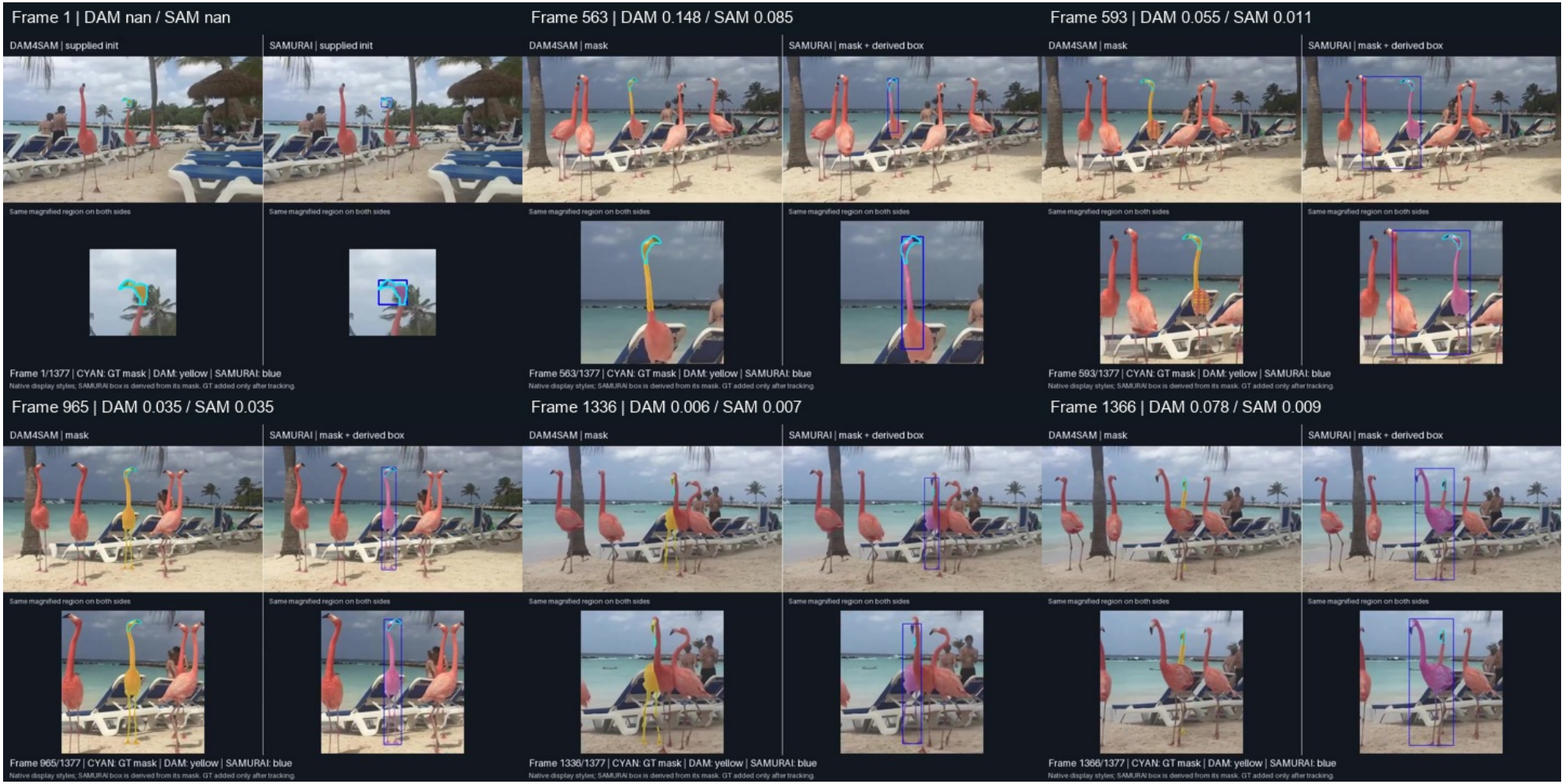
Cyan: active GT. Yellow: DAM4SAM. Blue: SAMURAI. The panels retain native mask predictions; the full dynamic sequence is available in the companion gallery.

Figure 5 | Confusion within a zebra group: LaSOT_zebra-16 / Key frames



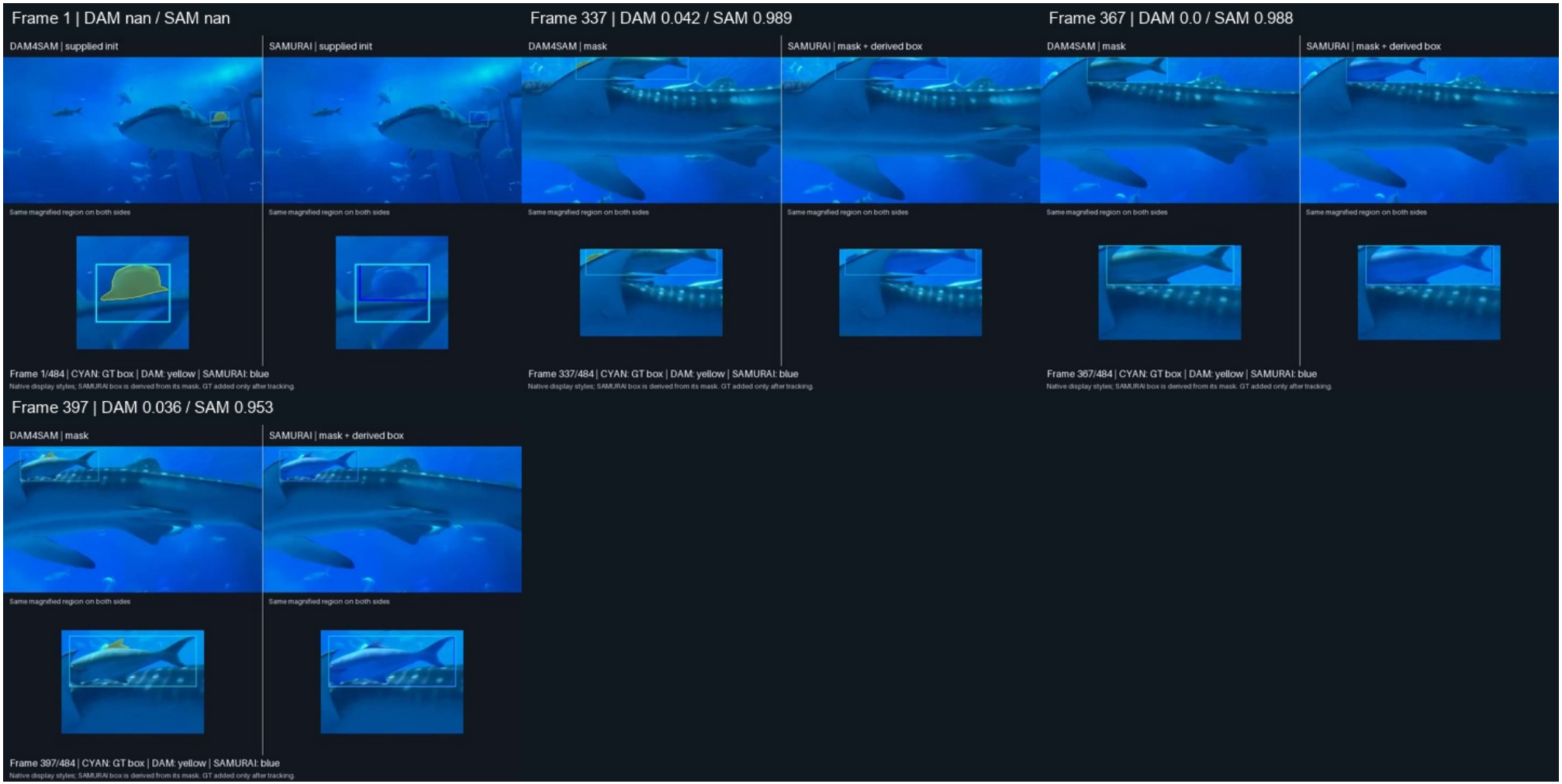
Cyan: active GT. Yellow: DAM4SAM. Blue: SAMURAI. The panels retain native mask predictions; the full dynamic sequence is available in the companion gallery.

Figure 6 | Low scores from expanding target extent: VOT2022-ST_flamingo1 / Key frames



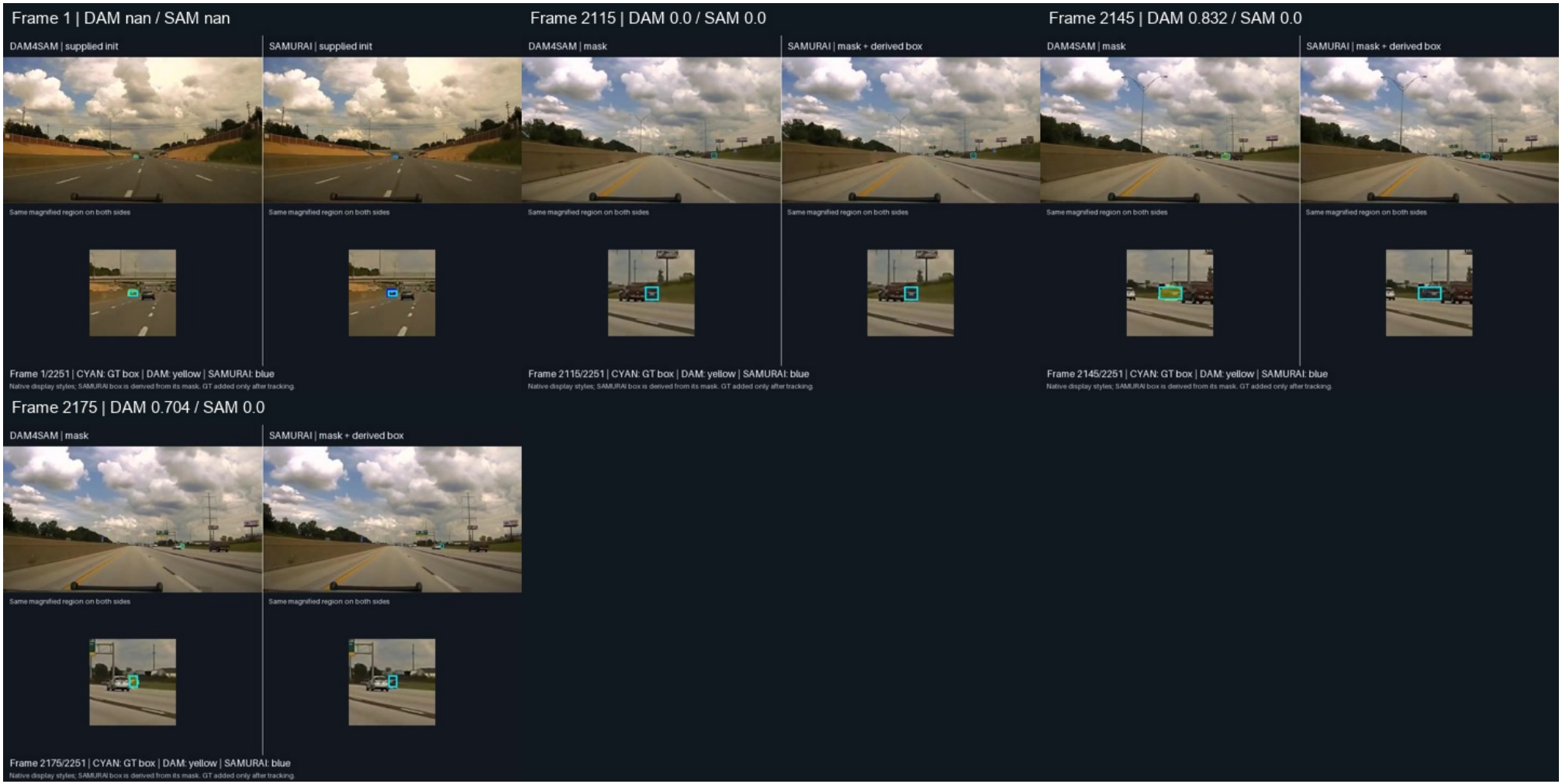
Cyan: active GT. Yellow: DAM4SAM. Blue: SAMURAI. The panels retain native mask predictions; the full dynamic sequence is available in the companion gallery.

Figure 7 | Control where SAMURAI is much better: UTB180_Video_0002 / Key frames



Cyan: active GT. Yellow: DAM4SAM. Blue: SAMURAI. The panels retain native mask predictions; the full dynamic sequence is available in the companion gallery.

Figure 8 | Divergence after valid annotation resumes: LaSOT_car-17 / Key frames



Cyan: active GT. Yellow: DAM4SAM. Blue: SAMURAI. The panels retain native mask predictions; the full dynamic sequence is available in the companion gallery.